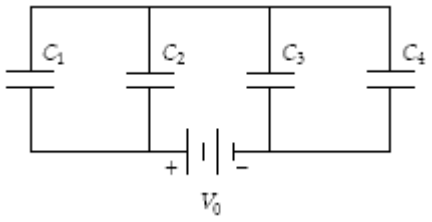
Physics 210 Tutorial #7 Multiple Choice Problems

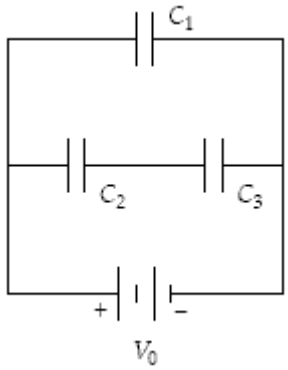
Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. What is the total energy stored by C_3 when C_1 = 50 μF, C_2 = 30 μF, C_3 = 36 μF, C_4 = 12 μF, and V_0 = 30 V?



a. 6.3 mJ
b. 25 mJ
c. 57 mJ
d. 1.6 mJ
e. 14 mJ

____ 2. What is the potential difference across C_2 when C_1 = 5.0 μF, C_2 = 15 μF, C_3 = 30 μF, and V_0 = 24 V?



a. 21 V
b. 19 V
c. 16 V
d. 24 V
e. 8.0 V

____ 3. A 6.0-μF capacitor charged to 50 V and a 4.0-μF capacitor charged to 34 V are connected to each other, with the two positive plates connected and the two negative plates connected. What is the total energy stored in the 6.0-μF capacitor at equilibrium?
a. 6.1 mJ
b. 5.7 mJ
c. 6.6 mJ
d. 7.0 mJ
e. 3.8 mJ

____ 4. A 25-μF capacitor charged to 50 V and a capacitor C charged to 20 V are connected to each other, with the two positive plates connected and the two negative plates connected. The final potential difference across the 25-μF capacitor is 36 V. What is the value of the capacitance of C?
   a. 43 μF
   b. 29 μF
   c. 22 μF
   d. 58 μF
   e. 63 μF

____ 5. A 30-μF capacitor is charged to 80 V and then connected across an initially uncharged capacitor of unknown capacitance C. If the final potential difference across the 30-μF capacitor is 20 V, determine C.
   a. 60 μF
   b. 75 μF
   c. 45 μF
   d. 90 μF
   e. 24 μF

____ 6. A parallel plate capacitor of capacitance C_0 has plates of area A with separation d between them. When it is connected to a battery of voltage V_0, it has charge of magnitude Q_0 on its plates. The plates are pulled apart to a separation $2d$ while the capacitor remains connected to the battery. After the plates are $2d$ apart, the capacitance of the capacitor and the magnitude of the charge on the plates are
   a. $\frac{1}{2}C_0$, $\frac{1}{2}Q_0$
   b. $\frac{1}{2}C_0$, Q_0
   c. C_0, Q_0
   d. $2C_0$, Q_0
   e. $2C_0$, $2Q_0$

____ 7. A parallel plate capacitor of capacitance C_0 has plates of area A with separation d between them. When it is connected to a battery of voltage V_0, it has charge of magnitude Q_0 on its plates. It is then disconnected from the battery and the space between the plates is filled with a material of dielectric constant 3. After the dielectric is added, the magnitudes of the charge on the plates and the potential difference between them are
   a. $\frac{1}{3}Q_0$, $\frac{1}{3}V_0$.
   b. Q_0, $\frac{1}{3}V_0$.
   c. Q_0, V_0.
   d. Q_0, $3V_0$.
   e. $3Q_0$, $3V_0$.

____ 8. A light bulb is rated at 30 W when operated at 120 V. How much charge enters (and leaves) the light bulb in 1.0 min?
   a. 17 C
   b. 15 C
   c. 14 C
   d. 13 C
   e. 60 C

____ 9. A wire (length = 2.0 m, diameter = 1.0 mm) has a resistance of 0.45Ω. What is the resistivity of the material used to make the wire?
   a. $5.6 \times 10^{-7}\ \Omega \cdot \text{m}$
   b. $1.2 \times 10^{-7}\ \Omega \cdot \text{m}$
   c. $1.8 \times 10^{-7}\ \Omega \cdot \text{m}$
   d. $2.3 \times 10^{-7}\ \Omega \cdot \text{m}$
   e. $7.1 \times 10^{-7}\ \Omega \cdot \text{m}$

____ 10. A conductor of radius r, length ℓ and resistivity ρ has resistance R. What is the new resistance if it is stretched to 4 times its original length?
   a. $\frac{1}{16}R$
   b. $\frac{1}{4}R$
   c. R
   d. $4R$
   e. $16R$

Answer Section

MULTIPLE CHOICE

1. ANS: A PTS: 1
2. ANS: C PTS: 1
3. ANS: B PTS: 1
4. ANS: C PTS: 1
5. ANS: D PTS: 1
6. ANS: A PTS: 1
7. ANS: B PTS: 1
8. ANS: B PTS: 1
9. ANS: C PTS: 1
10. ANS: E PTS: 1